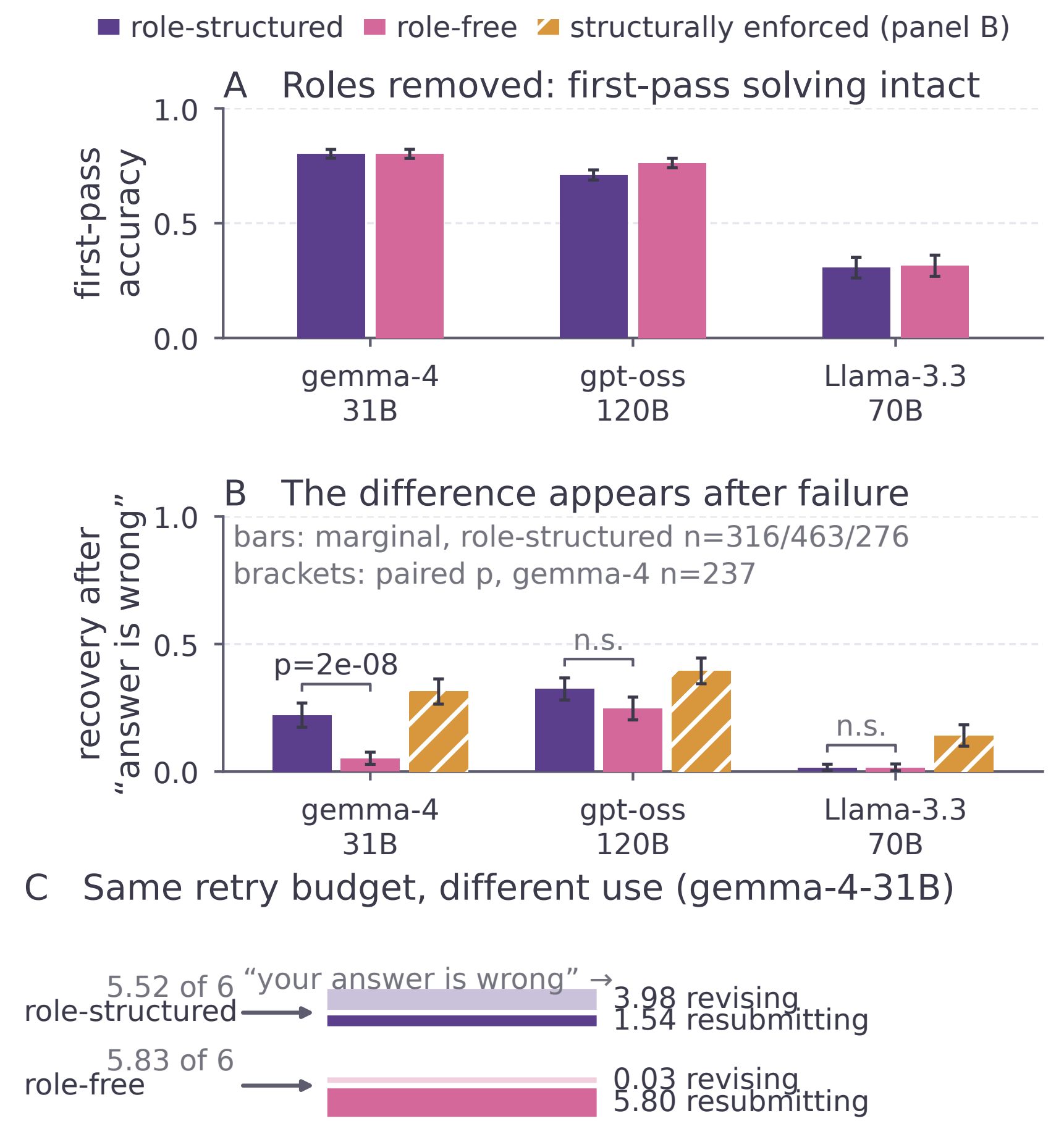
role-structured
role-free
structurally enforced (panel B)
A Roles removed: first-pass solving intact
first-pass accuracy
1.0
0.5
0.0
gemma-4 31B
gpt-oss 120B
Llama-3.3 70B
B The difference appears after failure
recovery after "answer is wrong"
bars: marginal, role-structured n=316/463/276
brackets: paired p, gemma-4 n=237
p=2e-08
n.s.
n.s.
C Same retry budget, different use (gemma-4-31B)
5.52 of 6
"your answer is wrong" →
role-structured
3.98 revising
1.54 resubmitting
5.83 of 6
role-free
0.03 revising
5.80 resubmitting